
Sample AWS CodeBuild Project – Step-by-Step

Scenario

Let's build a simple Node.js application using AWS CodeBuild. The app is stored in **GitHub**, and we'll:

- Pull the source from GitHub
 - Install dependencies
 - Run unit tests
 - Package the app
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Project Structure Example

Your GitHub repo might look like this:

```
my-node-app/  
├── buildspec.yml  
├── package.json  
├── index.js  
├── test/  
│   └── test.js
```

Step 1: Create **buildspec.yml**

This file tells CodeBuild what to do during the build process. Place it in the root of your repository.

version: 0.2

phases:

install:

runtime-versions:

nodejs: 18

commands:

- echo Installing dependencies...

- npm install

build:

commands:

- echo Running tests...

- npm test

artifacts:

files:

- '**/*'

You can modify the Node.js version and commands to fit your project.



Step 2: Create CodeBuild Project in Console

1. Go to the [AWS CodeBuild Console](#)
 2. Click “Create build project”
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Project Configuration

- **Project Name:** `my-nodejs-build`
- **Description:** *(optional)*
- **Source Provider:** `GitHub` (connect your GitHub account)
- **Repository:** Select your repository
- **Environment:**
 - Environment image: **Managed image**

- Operating system: `Amazon Linux 2`
 - Runtime: `Standard`
 - Image: `aws/codebuild/standard:7.0` (or latest)
 - Service Role: Create a new one or choose existing
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Buildspec

- Choose “**Use a buildspec file**”
 - This tells CodeBuild to look for `buildspec.yml` in your repo
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Artifacts

- **Type:** `Amazon S3`
 - **Bucket:** Choose or create an S3 bucket
 - **Name:** `output.zip` (optional)
 - **Artifact packaging:** `Zip`
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Logs

- Enable logs to **CloudWatch** for debugging (recommended)
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Finish

- Click “**Create build project**”
 - Once created, click “**Start build**”
 - Select the source version (e.g., **main** branch) and start the build
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Step 3: View Build Results

- Monitor progress in **Build history**
 - View **phases**: install, build, post-build
 - Check **logs** in CloudWatch for details
 - If successful, output artifacts will be in your S3 bucket
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Optional Enhancements

- **Trigger builds automatically** on GitHub pushes (via webhook)
 - **Add CodePipeline** to deploy automatically after build
 - **Use environment variables** to parameterize builds
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Would you like:

- A **sample GitHub repo** for this?
- Instructions for **Docker builds** or **Lambda deployment**?
- A **Terraform/IaC version** to automate setup?